CS60092: Information Retrieval

ReARTeR: Trustworthy Reasoning in RAG Systems

A rigorous reproduction study of Process Reward Models for Retrieval-Augmented Generation — examining architecture, experimental results, critical limitations, and novel engineering improvements.

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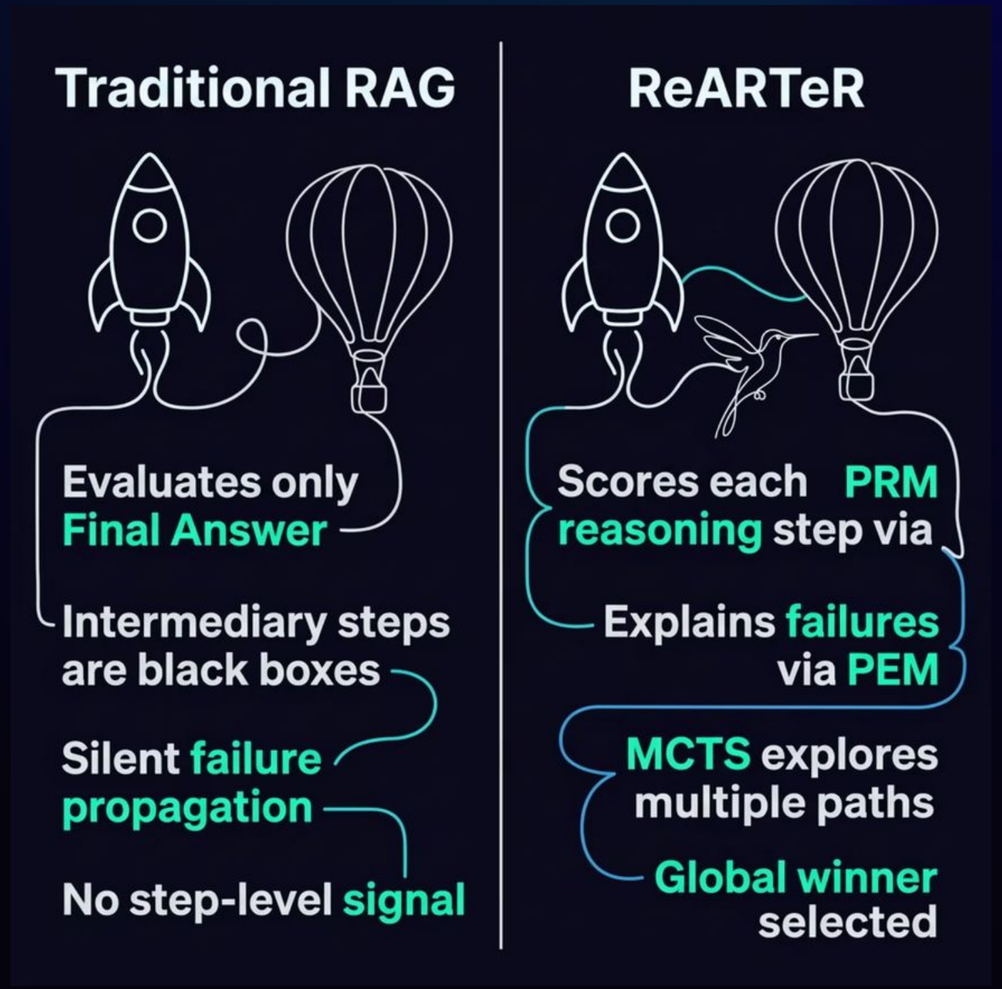
THE PROBLEM

RAG Systems Lack Reasoning Quality Signals

Existing RAG evaluates only the **Final Answer**, leaving intermediate reasoning steps as **black boxes**.

- **Outcome-only feedback:** Silent corruption of the reasoning chain goes undetected
- **Unverified multi-hop steps:** Dependencies fail silently when Step 1 is hallucinated
- **Retained noise:** FLARE-style retrieval decisions don't guarantee improved logic



METHODOLOGY

ReARTeR: PRM & PEM Integration

ReARTeR combines three components in an MCTS loop to explore and score reasoning paths.

Reasoning Pipeline

Generates follow-up queries via BM25 retrieval across multi-hop reasoning chains

PRM (Process Reward Model)

7B LLM scoring each reasoning step on a 0–1 scale for correctness

PEM (Process Explanation Model)

Generates natural language "Why" explanations for step-level failures

Stack: RTX 3060 · Qwen 2.5 (3B) via Ollama · FlashRAG Framework

REPRODUCED RESULTS

Experimental Setup & Pilot Reproductions

Dataset & Pipeline

Synthetic 192-doc corpus, 110 QA pairs across 7 domains (Geography to Food). MCTS explores multiple reasoning paths; PRM selects the global winner.

Engineering Contributions

Fixed **4 critical FlashRAG bugs**: pdb breakpoints, serialization errors, and tokenizer fallbacks — enabling stable reproduction runs.

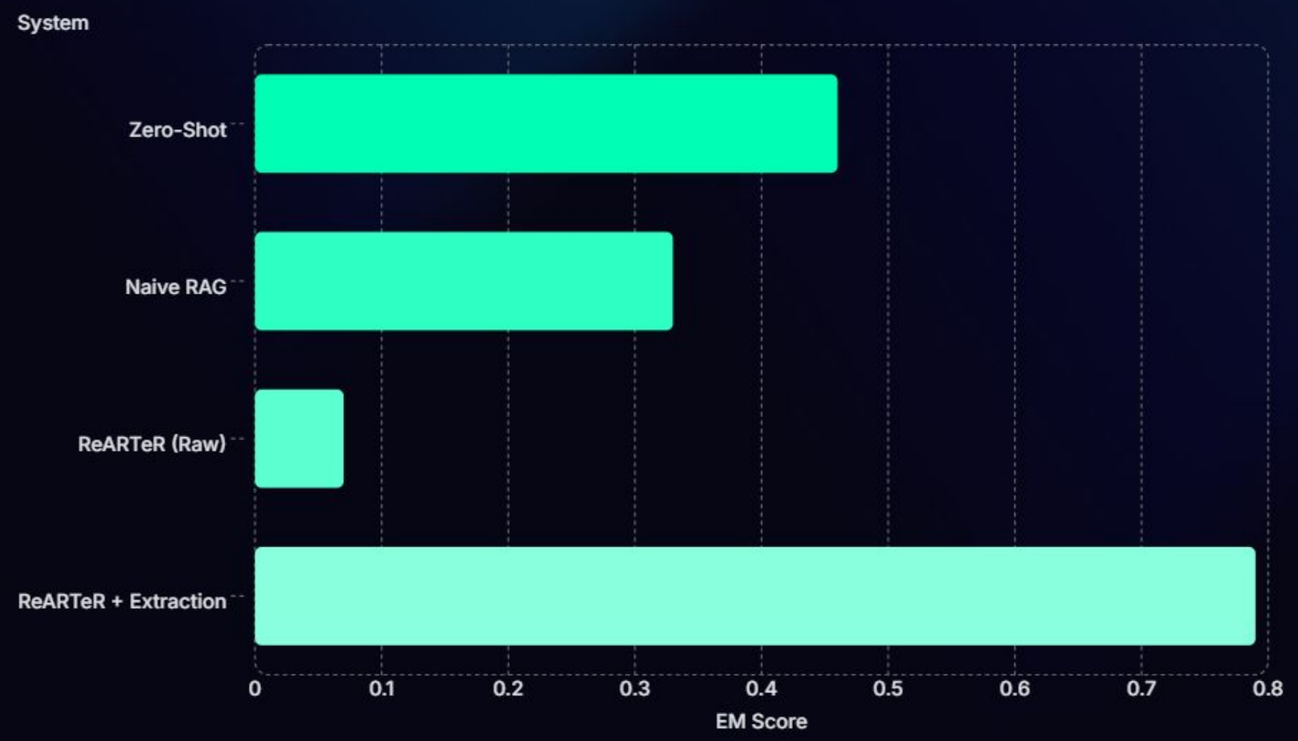
Pilot Metrics (15 Questions)

System	EM ↑	F1 ↑	Accuracy ↑	Recall ↑
Zero-Shot	0.200	0.341	0.800	0.733
Naive RAG	**0.267**	**0.467**	**0.867**	—
ReARTeR (no PRM)	0.067	0.226	0.733	0.800

The EM Gap: Exact Match penalises correct but verbose reasoning — e.g., "The Himalayas" vs "Everest is in the Himalayan range..."

REPRODUCED RESULTS

Performance Analysis Across Scale (50–500 Questions)



Key Findings

- ReARTeR is **8× slower** than Zero-Shot due to iterative reasoning overhead
- Zero-Shot beats Naive RAG on synthetic data
- ReARTeR maintains high **semantic accuracy (0.85–0.90)** throughout scaling

Core conclusion: PRM is **essential**, not optional — the raw pipeline validates the paper's central claim.

LIMITATIONS

Critical Gaps & Bottlenecks

⚠️ Weight Ghosting

PRM/PEM weights were never released — core results are **irreproducible** by the community

📏 Metric Bias

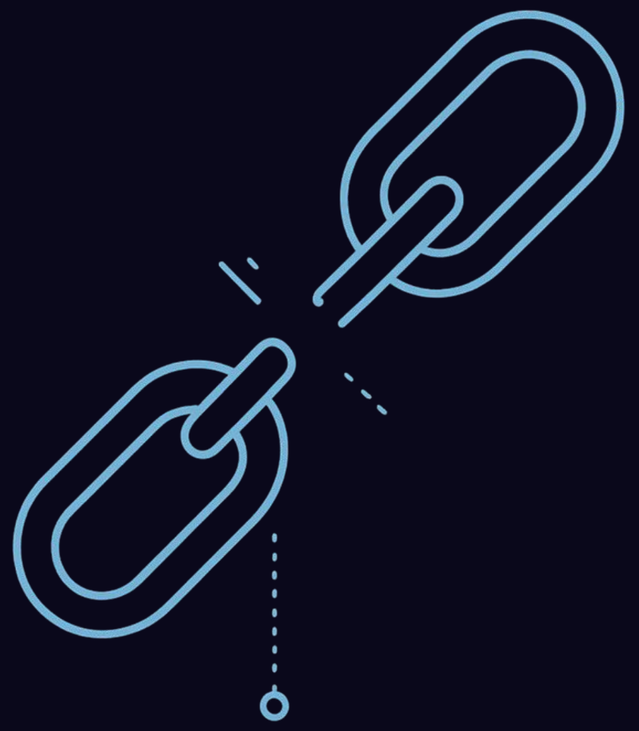
EM is outdated for reasoning-centric RAG; penalises semantically correct but verbose outputs

🖥️ Hardware Ceiling

Generator + PRM (7B+7B) exceeds consumer hardware limits (12GB VRAM)

🔬 Small Model Failure

3B models struggle with the rigid structured prompting the paper requires



PROPOSED IMPROVEMENTS

Engineering a Better ReARTeR

Post-Processing Extraction

A 20-token LLM call extracts clean answers, resolving the EM flaw by isolating the core response from verbose reasoning

LLM-as-Judge

Self-scoring prompts replace missing PRM weights, enabling step-level quality assessment without proprietary model access

Uncertainty-Adaptive Retrieval

Retrieve only when the model signals uncertainty (e.g., "I'm not sure"), drastically cutting unnecessary calls

Query Reformulation

Rewrite questions into optimised BM25 keywords for sharper context retrieval and reduced noise

RESULTS

Post Mid-Sem Achievements & Novel Contributions

0.790

Best EM Score

ReARTeR + LLM Extraction — highest across all baselines

0.845

F1 Score

ReARTeR + Extraction, confirming precision and recall balance

+0.720

EM Lift

Raw ReARTeR (0.07) → Extraction layer (0.79)

1%

Retrieval Rate

Uncertainty-Adaptive RAG achieved 0.61 EM retrieving only 1% of the time

System	EM	F1
Zero-Shot	0.46	—
Naive RAG	0.33	—
ReARTeR + Extraction (Winner)	**0.79**	**0.845**

Final Finding: ReARTeR was always semantically accurate — our extraction layer proved it by lifting EM from 0.07 to 0.79.

Thank You